

Women Safety & Support Application

Project Adhira — A Unified Ecosystem for Emergency Response, AI Legal Guidance & Community Care

EMERGENCY SOS Instant GPS & SMS Alert	ADHIRA AI Legal & Crisis Assistant	VERIFIED VOLUNTEERS Voter ID & NGO Vetted	SAFE FORUM Zero-Shot AI Moderation
---	--	---	--

Target Operating Systems: Web, Windows, Android, iOS	Release State: Stable Production v1.0.0+1 (Oct 2026)
Frontend Framework: Flutter 3 (Material Design 3)	Backend Architecture: Node.js, Express 5, MongoDB
AI Infrastructure: Groq (LLaMA 3.1 8B) & Google Gemini	Security & Auth: JWT Auth, Bcrypt, Voter ID Verification

Executive Mission & Solution Overview

Executive Summary: Project Adhira bridges acute emergency response with sustained post-crisis support. Unlike conventional safety apps that only offer a raw panic buzzer, Adhira delivers an integrated continuum of care: one-tap satellite GPS tracking and cellular SMS broadcast, an on-demand verified volunteer network, an AI-moderated community feed, and 'Adhira'—a 24/7 empathetic conversational companion specialized in Indian legal statutes (IPC & Bharatiya Nyaya Sanhita - BNS) and crisis de-escalation in regional languages. Built on Flutter and Express 5, the platform ensures privacy, zero-toxicity auto-quarantine, and verified human intervention.

DOCUMENT OUTLINE & ROADMAP	
1. The Core Problem & The 4 Architectural Pillars	Systemic safety gaps, care continuum, demographic reach
2. End-to-End System Architecture & Tech Stack	Flutter client, Express 5 API, MongoDB, Groq & Gemini services
3. Emergency SOS & Telemetric Dispatch Engine	Satellite GPS, native cellular SMS intent, 112/1091 dialers
4. 'Adhira' AI Legal & Safety Conversational Companion	LLaMA 3.1 8B, Gemini failover, IPC/BNS rights, empathetic tone
5. Community Feed & Verified Volunteer Network	Zero-shot AI moderation, Voter ID vetting, 1-on-1 private rooms
6. Database Schemas, Security & Quick Start Guide	MongoDB models, privacy rules, PowerShell launcher, v2.0 roadmap

1. The Core Problem & The Four Pillars

Women frequently encounter acute physical vulnerabilities during commutes, at workplaces, and within domestic spheres. Standard safety applications fail to address these realities due to four critical architectural flaws:

- **Single-Point Failure of 'Panic Buzzers':** Conventional SOS apps only transmit a raw coordinate text without providing guidance on what to do during or after the incident.
- **Severe Legal Information Gap:** Victims of stalking, domestic harassment, or cyberbullying often do not know that Indian law mandates *Zero FIR*, or how to invoke Section 354 IPC / BNS protections.
- **Lack of Trusted Civilian Intermediaries:** Victims hesitate to approach unverified strangers online without verified credentials, NGO affiliations, or peer ratings.
- **Isolation & Retaliation Fears:** Survivors of abuse fear retaliation if they post publicly, lacking secure spaces to share anonymously without toxicity.

The Four Architectural Pillars of Adhira:

Pillar 1: Emergency SOS Telematics
Immediate satellite GPS coordinate capture, persistent cloud audit logging, native SMS broadcast with direct Google Maps hyperlinks to trusted contacts, and 112/1091 dialer escalation.

Pillar 2: 'Adhira' AI Legal & Safety Guide
Sub-400ms conversational LLM (Groq LLaMA 3.1 8B with Gemini failover). Educates users on Indian legal statutes, de-escalates panic, and maintains an empathetic sisterly tone.

Pillar 3: Verified Volunteer Network
Govt. Voter ID verified civilian and NGO volunteers. Features real-time Active/Inactive availability switches, skill matching, and 1-on-1 private guidance rooms.

Pillar 4: Safe Community & Grievance AI
Anonymous story sharing with zero-shot Gemini multimodal AI moderation: classifies urgency (green/yellow/red), tags domain categories, and auto-quarantines toxic content.

Target Demographics & Operational Impact Matrix:

User Demographic	Key Vulnerabilities Faced	Platform Solutions Provided
Solo Commuters & Students	Late-night travel, unfamiliar transit routes, cab harassment, street stalking risks.	One-tap SOS, live Google Maps SMS broadcast, trusted emergency contact circle.
Domestic Abuse Survivors	Coercive control, physical isolation, lack of legal rights awareness.	Adhira legal advisor (PWDVA, IPC 498A), verified NGO volunteer private sessions.
Workplace & Cyber Victims	Stalking, online defamation, workplace harassment under POSH Act.	Anonymous community posting, AI grievance classification, legal rights guidance.
Social Workers & Volunteers	Coordination bottlenecks, lack of structured verification, volunteer burnout.	Availability switch, ticket triage dashboard, verified badges, rating metrics.

2. System Architecture & Technology Stack

Project Adhira is engineered on a resilient, decoupled client-server architecture. A single, responsive Flutter codebase serves Web, Windows Desktop, and Mobile clients, connecting to a high-throughput Express 5 REST API gateway and intelligent cloud AI microservices.

CROSS-PLATFORM FRONTEND CLIENT (Flutter 3.x / Dart) Responsive User Shell • Volunteer Dashboard • Material 3 UI • Provider Polling • Device GPS Telematics			
▼ HTTPS / Secure REST API Gateway (JSON) ▼			
BACKEND MICRO-SERVICES & BUSINESS LOGIC (Node.js & Express 5.2) JWT Role Guard • SOS Dispatcher • Volunteer Triage • Session Router • Grievance Controller			
MongoDB Store Users, SOS Logs, Posts, Chat Sessions	Groq AI Engine LLaMA 3.1 8B Sub-400ms Inference	Google Gemini Zero-Shot Grievance & Web Grounding	Device Hardware GPS Satellites & Native SMS Telephony

Technology Specifications & Dependency Matrix:

Component Layer	Libraries & Frameworks	Key Responsibilities
Frontend Client	- Flutter 3.2.0+ (Dart) - provider: ^6.1.2 - geolocator: ^12.0.0 - url_launcher: ^6.3.2 - image_picker: ^1.1.2	Cross-platform responsive shells for Seekers and Volunteers. Manages background notification polling, device GPS interrogation, and telephony SMS triggers.
Backend Server	- Node.js & Express 5.2.1 - mongoose: ^9.4.1 - jsonwebtoken: ^9.0.3 - bcryptjs: ^3.0.3 - cors: ^2.8.6	Stateless REST API gateway. Implements password encryption, JWT issuance, role-based route guardrails, 50MB media payload handling, and AI request orchestration.
Database	- MongoDB 6+ / Atlas - Docker Compose container - Mongoose ODM	Stores schemas for Users, Emergency Contacts, Verified Volunteers, SOS Events, Community Posts, Comments, and 1-on-1 Chat Sessions.
Cloud AI Services	- Groq Cloud (LLaMA 3.1 8B) - Google Gemini Flash & 2.5 - Zero-shot Sentiment Engine - Gemini Search Grounding	Sub-second conversational generation for Adhira assistant; zero-shot multimodal grievance & toxicity classification; live search for safety articles.

3. Emergency SOS & Telemetric Dispatch Engine

The Emergency SOS subsystem is designed for deterministic execution under acute danger. When activated, the system executes a 4-phase parallel response that combines satellite telemetry, immutable cloud incident logging, and offline-resilient cellular SMS broadcast.

Response Phase	Action Taken	Technical & Operational Mechanism
Phase 1: Satellite Geocoding	Hardware GPS Interrogation	Queries device GPS hardware using <code>Geolocator.getCurrentPosition()</code> with high accuracy. Extracts latitude and longitude in milliseconds.
Phase 2: Cloud Incident Logging	Secure POST to <code>/api/sos</code>	Transmits coordinates, timestamp, and distress notes to MongoDB. Creates a permanent forensic record for responder auditing.
Phase 3: Cellular SMS Dispatch	Native OS SMS Broadcast	Compiles all registered emergency contacts into comma-separated recipient strings. Sends a direct Google Maps link: <code>https://maps.google.com/?q={lat},{lng}</code> via native cellular radio even if mobile data is lost.
Phase 4: Hotline Escalation	Telephony Shortcuts	Surfaces immediate one-tap dialers for national emergency helplines: 112 (Unified National Emergency) and 1091 (Women Helpline).

Emergency Circles & Offline Cellular Resilience Rationale:

Users can configure an unlimited list of trusted emergency contacts (name and phone number) in their profile. In crisis situations, mobile data (4G/5G) often drops in basements, transit tunnels, or remote roads. By packaging the distress message and Google Maps location into a standard **telephony SMS intent**, Project Adhira ensures the emergency beacon is dispatched through base cellular signaling without depending on internet access.

Live Telemetric SMS Dispatch Format

Automated Cellular Distress SMS (Delivered to Emergency Contacts):

■ **SOS EMERGENCY!** I need help. My location: `https://maps.google.com/?q=28.6139,77.2090`. Feeling unsafe near Central Metro station."

- **Direct Action:** Recipients can tap the link to open live turn-by-turn navigation directly to the user's coordinates.
- **Zero Internet Reliance:** The SMS is transmitted via native cellular baseband even when WiFi and mobile data are disconnected.

Forensic Reliability & Legal Protection

Forensic Incident Audit Trail: Every SOS trigger creates an immutable entry in MongoDB. This chronological record stores exact GPS coordinates, timestamps, and notes, providing formal documentation that can be presented to police investigators or legal counsel to establish the timeline of an assault or stalking event.

4. 'Adhira' Intelligent AI Safety & Legal Assistant

Adhira is an autonomous AI conversational companion built into the application. Configured with a distinct persona of a *'caring elder sister who knows the law'*, Adhira de-escalates panic, explains statutory rights in simple everyday language, and provides step-by-step guidance without legal jargon.

Resilient Dual-Engine AI Cascade:

To ensure 99.9% uptime during emergencies, the chatbot backend implements an intelligent fallback cascade:

- **Primary Engine (Groq LLaMA 3.1 8B):** Delivers blazing sub-400ms inference with low temperature (0.3) for concise, supportive, and factually grounded advice.
- **Secondary Engine (Google Gemini Flash):** Automatically engaged if Groq hits rate limits, network outages, or key exhaustion.
- **Tertiary Deterministic Net:** Built-in keyword heuristics return pre-compiled helpline actions if all cloud services fail.

Statutory Protections & Indian Legal Knowledge Base:

Statutory Protection	Key Provisions Covered	How Adhira Advises the User
Zero FIR Mandate	Sec 154 CrPC / BNSS provisions	Clarifies that any police station must register an FIR for cognizable crimes regardless of jurisdiction, without redirecting the victim.
Molestation & Stalking	IPC Sec 354, 354A, 354D / BNS Equiv.	Explains elements required to prove electronic/physical stalking, evidence gathering steps (call logs, screenshots), and non-bailable clauses.
Domestic Violence Act	PWDVA 2005 & IPC Sec 498A (Cruelty)	Outlines immediate Protection Orders, Right to Reside in shared households, and how to reach Protection Officers and recognized NGOs.
Cyber Crimes & Anonymity	IT Act Sec 66E, 67, 67A & Helpline 1930	Provides clear instructions on filing complaints for non-consensual image sharing, online blackmail, and reporting to National Cyber Crime portal.

Conversational Persona & Anti-Victim Blaming Guardrail

Sample Conversation with Adhira (Empathetic & Legal Guidance):

User: "Someone has been following me from the metro station for the last 3 days and taking photos. What can I do?"

Adhira: "I hear you, and you are not alone. First, please ensure you are in a crowded, well-lit place right now. What this person is doing is a serious crime under Section 354D of the Indian Penal Code (Stalking). You have the right to file an FIR at any police station without delay (Zero FIR). Here are three immediate steps: 1) Call 112 or share your live location with a trusted contact. 2) Note down their vehicle number or appearance safely. 3) You can also connect with one of our verified legal volunteers right here in the app. Stay strong, I am here with you. ■"

5. Community Forum & Verified Volunteer Network

Project Adhira's Community & Guidance subsystems transform isolated victims into supported community members. Every post on the public feed is monitored by **Zero-Shot Multimodal Gemini AI**, while the Volunteer ecosystem connects seekers to vetted human advocates.

AI Metric	Triage Level	System Action & Feed Behavior
Urgency Level	Green (Normal)	General stories, advice, safety questions. Rendered with green badge.
	Yellow (Distress)	Emotional pain, anxiety, seeking advice. Highlighted to volunteers for rapid support.
	Red (Crisis)	Active physical abuse, severe danger. Triggers automated prompt to user with SOS button.
Toxicity Moderation	High Toxicity	Auto-Quarantine: Immediately flagged, soft-deleted (isDeleted: true), and hidden with notice: <i>'Deleted due to unethical content'</i> .
Domain Tagging	Domain Tags	Auto-classified: <i>Harassment, Domestic Violence, Stalking, Mental Health, Workplace Safety.</i>

Verified Volunteer Lifecycle & 1-on-1 Private Sessions:

Verified Volunteer Ecosystem & Accountability

- **Voter ID Verification:** Volunteers submit Govt. Voter ID credentials, phone, address, and NGO affiliations. Verified profiles display a verified badge.
- **Availability Toggle:** Volunteers can toggle between **Active** and **Inactive** status. Only active volunteers appear in urgent search suggestions.
- **1-on-1 Private Chat Sessions:** Seekers can request private guidance from a specific volunteer. When accepted, a private encrypted chat room is spun up.
- **Ratings & Accountability:** Users submit 1-5 star ratings, written reviews, hours spent, and people helped, building a transparent track record.

Companion Safety Apps Directory & Live AI Blogs:

The app includes a curated directory linking directly to specialized safety utilities: *bSafe* (live tracking), *Raksha* (hardware volume SOS), *Himmat* (police direct video dispatch), *Circle of 6* (discreet group alerts), *My Safetipin* (street lighting safety audits), and *Shake2Safety* (haptic shake trigger). Additionally, the 'Know Your Community' tab pulls live, authentic blogs on women's legal rights and self-defense using Gemini Google Search Grounding.

6. Database Schemas, Security & Quick Start Guide

Core Database Schemas (MongoDB / Mongoose):

Collection	Key Attributes & Schema Fields	Operational Function
User	name, email, password, googleId, role ('user' 'volunteer'), voterIdVerified, volunteerAvailability, emergencyContacts, ratings, followers, following	Stores profile credentials, emergency contacts, volunteer ratings, and mutual follow permissions.
HelpRequest	requesterId, volunteerId, sosId, message, assistanceNote, hoursSpent, peopleHelped, rating (1-5), status ('pending' 'accepted' 'completed')	Tracks guidance ticket lifecycle, assigned volunteers, and resolution metrics.
Post	userId, content, mediaUrl, mediaType ('text' 'image' 'audio'), isAnonymous, distressLevel, urgencyColor, toxicityLevel, isDeleted, likes	Community feed entries with AI sentiment tags and soft-delete quarantine flags.
PrivateChat	Session: { helpSeekerId, volunteerId, status } Message: { sessionId, senderId, content, urgencyColor, toxicityLevel }	1-on-1 private messaging exchange between seeker and volunteer.
SOS Logs	userId, location: { lat: Number, lng: Number }, notes, timestamps	Permanent historical audit log of emergency incidents.

Security, Privacy & Data Minimization:

- **Anonymous Identity:** Survivors can post and seek help anonymously; real identity fields are scrubbed from feed payloads.
- **Zero-Exposure Profiles:** Passwords (salted Bcrypt), voter IDs, and emergency contact phones are strictly omitted from public API queries.
- **Follow Request System:** Prevents unwanted stalking between platform users by requiring explicit approval before profile visibility.

Quick Start & Execution Guide (PowerShell):

```
# 1. Quick Automated Launch: powershell -ExecutionPolicy Bypass -File .\run-windows.ps1 -Port 5000
# 2. Run Backend Manually: cd Backend && npm install && $env:PORT=5000 && npm run dev
# 3. Seed Default Admin (admin/admin): cd Backend && npm run seed:admin
# 4. Run Flutter Client (Web): cd frontend && flutter pub get && flutter run -d chrome
--dart-define=API_BASE_URL=http://127.0.0.1:5000
```

Future Strategic Roadmap (v2.0):

- **Acoustic Distress & Scream Detection:** On-device ML listening in background for scream acoustics or distress voice triggers.
- **BLE Mesh Networking:** Peer-to-peer relaying of SOS beacons across nearby devices even when cell data is completely disabled.
- **Police ERSS 112 CAD Integration:** Direct API dispatch into State Police Computer-Aided Dispatch networks for immediate cruiser dispatch.